Attendance

1. Real-World Teacher Workflow

Typical classroom process:

1. Teacher opens class attendance.
2. System shows **list of all students in the class**.
3. All students are assumed **present by default**.
4. Teacher marks students who are **absent or late**.
5. Teacher clicks **Submit**.

Total time:

 10–30 seconds

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2. Best Attendance UI for Teachers

The system should open directly to **today's class attendance page**.

Example screen:

```
Grade 5 - Mathematics
Teacher: John Kamau
Date: 12 March 2026
Time: 08:05 AM
```

Student list:

```
☑ Brian Mwangi
☑ Mercy Wanjiku
☑ Kelvin Otieno
☐ David Mutua (Absent)
☑ Aisha Hassan
```

Legend:

```
☑ Present (default)
☐ Absent
 Late
 Excused
```

Teacher simply **ticks absent students**.

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3. Default Behavior (Important)

System automatically marks:

```
All students = Present
```

Teacher only changes:

- absent
- late
- excused

This drastically **reduces teacher workload**.

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4. Auto-Filled Data

The system should automatically capture:

Field	Auto Generated
Date	system date
Time	system timestamp
Teacher	logged-in user
Class	teacher assignment
Subject	timetable

Teacher **should not type these**.

Example database record:

```
attendance_date: 2026-03-12
teacher_id: 104
class_id: grade5
subject: mathematics
recorded_time: 08:05
```

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5. Attendance Status Types

Your system should support multiple attendance statuses.

Recommended statuses:

Status	Meaning
Present	default
Absent	not in school
Late	arrived late
Excused	approved absence
Sick	medical absence

Teachers usually only select:

```
Absent
Late
```

Other statuses can be updated by **admin or office staff**.

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6. Speed Features (Very Important)

To improve usability:

Search Student

Teacher can search if class is large.

Quick Mark All Absent

Rare but useful.
or teacher can read names as they mark present or absent

Example:

```
Mark all absent
Mark all present
```

Bulk Edit

Example:

```
Select multiple students → Mark Late
```

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7. Attendance via Timetable (Best UX)

Teachers should not navigate classes manually.

System should show **today's timetable**.

Example dashboard:

```
Today's Classes

08:00 Math → Take Attendance
09:00 English → Take Attendance
10:00 Science → Take Attendance
```

Teacher clicks:

```
Take Attendance
```

Student list opens.

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8. Prevent Duplicate Attendance

System should prevent teachers from taking attendance **twice**.

Example message:

```
Attendance already recorded for this class.
```

Teacher can only **edit existing attendance**.

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9. Late Attendance Lock (Optional)

Schools often lock attendance after a certain time.

Example:

```
Attendance locked after 10:00 AM
```

Only admin can edit afterwards.

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10. Parent Notifications

Attendance should trigger alerts.

Example:

```
Brian Mwangi was marked absent today.
```

Send to:

- parent app
- SMS
- email

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11. Attendance Dashboard for Teachers

Teachers should see **attendance summary**.

Example:

```
Class Attendance Today

Present: 28
Absent: 2
Late: 1
```

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12. Admin Attendance Monitoring

Admin should see **school-wide attendance dashboard**.

Example:

```
School Attendance Today

Students Present: 920
Absent: 32
Late: 15
Attendance Rate: 96.6%
```

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13. Attendance Reports

System should generate reports.

Examples:

Student Report

```
Brian Mwangi

Present: 120 days
Absent: 6 days
Late: 2 days
```

Class Report

```
Grade 5

Average Attendance: 94%
```

School Report

```
Monthly Attendance Rate
```

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14. Advanced Attendance Methods

Your system can support multiple methods.

Manual (Most common)

Teacher marks attendance.

QR Code

Students scan a code when entering class.

RFID Cards

Students tap card.

Biometric

Fingerprint scanners.

For your SaaS launch, **manual attendance is enough**.

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15. Handling Late Students

Teacher marks student:

```
Late
```

System records:

```
arrival_time
```

Example:

```
Late by 10 minutes
```

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16. Absence Reason Workflow

If student is absent:

Parent can later submit reason.

Example:

```
Medical appointment
```

Admin can update:

```
Absent → Excused
```

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17. Database Structure

Table:

```
attendance_records
```

Fields:

```
id
student_id
class_id
subject_id
teacher_id
attendance_date
status
recorded_time
notes
```

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18. Performance Optimization

Attendance must be **fast even with large classes**.

Techniques:

- preload student list
- save with single API request
- auto-save option

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19. Best Mobile Design

Teachers may use phones or tablets.

Mobile layout:

```
Student Name     Status

Brian Mwangi     [Present]
Mercy Wanjiku    [Absent]
Kelvin Otieno    [Present]
```

One tap changes status.

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20. Ideal Workflow for Your SMS

Final teacher flow:

```
Login
 ↓
Dashboard
 ↓
Today's Class
 ↓
Open Attendance
 ↓
Tick Absent Students
 ↓
Submit
```

Total time:

 Under 15 seconds